

Alpha Foundry User Guide

How to operate the tool, from first start to alphas that pass on WorldQuant BRAIN

This guide explains how to install and operate Alpha Foundry: building the data, connecting to WorldQuant BRAIN, generating alphas in the Studio, the Idea Forge and the Miner, reading the quality grade, and taking candidates through BRAIN to submission.

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1. What Alpha Foundry is

Alpha Foundry is a workbench for building WorldQuant BRAIN alphas. You write, generate and refine Fast Expression alphas, and the tool tells you which ones are likely to pass on BRAIN. Everything runs in your browser at <http://127.0.0.1:8765>, served by a program on your own computer.

It has two engines that work together:

Engine	What it is	What it is for
Local simulator	A BRAIN-like backtester running on free data (Yahoo Finance prices, SEC EDGAR fundamentals)	Fast screening: thousands of candidates per hour, offline, at no BRAIN cost
BRAIN connection (optional)	Your BRAIN account, reached through BRAIN's API	The ground truth: real metrics, real checks, real self- and production-correlation

The local simulator is a **proxy**. Its data, universe and in-sample window differ from BRAIN's, so the tool never treats a bare local pass as good enough. Every alpha gets a **quality grade** (A to D). Grade A asks for a clear margin above BRAIN's thresholds plus evidence that the edge is real. The best candidates then go to BRAIN for the final verdict.

The workflow at a glance

1. Build real data (once, then update now and then).
2. Generate candidates: write them in the **Studio**, describe an idea in the **Idea Forge**, or let the **Miner** search.
3. Keep the grade A and B alphas.
4. Test them on BRAIN (**Run on BRAIN**), or paste the exports into BRAIN yourself.
5. Submit the ones BRAIN marks ready, on the BRAIN website.
6. Sync the results back so the tool learns what passes.

Important. Alpha Foundry never submits an alpha for you. Submission is always a manual click on the BRAIN website.

2. Installing and starting the tool

Requirements

- Windows 10 or 11
- [uv](#) (Python package manager). Python 3.13 is fetched automatically.
- Node.js 20 or newer. It is only needed to build the user interface, which the start script does for you.
- About 3–4 GB of free disk space for the broad data pool, 1–2 GB for the S&P 1500 pool

Starting

Double-click **start.bat** in the project folder. The script:

1. installs or updates the Python environment (**uv sync**);
2. builds the web interface if its source changed (**npm run build**);
3. on the very first run only, compiles the simulation engine (1–3 minutes);
4. starts the server and opens <http://127.0.0.1:8765> in your browser.

Keep the console window open while you work. Closing it, or pressing **Ctrl+C** in it, stops the tool. Background jobs stop with it, but everything already saved stays saved.

Options

- `start.ps1 -NoBrowser` starts without opening a browser tab.
- `start.ps1 -Port 8800` uses another port. If 8765 is busy, the server picks the next free port and prints it.
- Manual start: open a terminal in `backend` and run `uv run alphafoundry serve --open`.

The first time you open it

The tool starts on a small **synthetic demo dataset** (300 made-up stocks) so the screens are not empty. A warning banner stays at the top while the demo data is active. **Mining is disabled on demo data**, for a good reason: the demo generator plants value, quality, reversal and momentum effects on purpose. Alphas that look excellent there (local Sharpe above 2) do not exist on BRAIN. Your next step is always section 3.

3. Building real data (do this first)

Open **Data** in the sidebar.

Step by step

1. **Contact email.** Enter your email and press **Save**. SEC EDGAR and Wikipedia require a contact address in the User-Agent of automated downloads. The email is sent to those two sites only, and only for that purpose.
2. **Stock pool.** Choose one:

Pool	What it contains	Universes	First build	Disk
Broad US (recommended)	Every listed US common stock from SEC's exchange list, kept when it ranks among the ~3,450 most liquid at any month end	TOP100 ... TOP3000, like BRAIN's USA region	40–70 min	2–3 GB
S&P 1500	Current S&P 500, 400 and 600 constituents	TOP100 ... TOP1500	10–20 min	~1 GB

3. Optional: fill **max tickers** for a quick trial build (for example 200). Leave it empty for the full build.
4. Press **Build real data**. The progress bar shows the phases: universe → prices → classification (broad pool only) → fundamentals → assemble.

When the build finishes, the tool switches to the real data automatically. The header chip changes from *Demo data* to *Real data*, and mining becomes available.

Good to know

- **Pause OneDrive** while the build runs. The project folder is synced, and a 2–3 GB build can otherwise upload files while they are being written.
- **Updates are incremental.** Press **Update data** (the same button) now and then, for example weekly. Only new prices and changed filings are downloaded.
- **Yahoo rate limits (HTTP 429)** make the downloader wait and retry. If a build stops, run it again; cached tickers are skipped.
- **The broad pool is about twice as slow to simulate** as the S&P 1500 pool, because it holds more than twice as many stocks. It is much closer to what BRAIN tests, which is the point.

How the data is made honest

- **Point in time.** A fundamental value becomes usable on the trading day **after** its SEC filing date. The first-filed figure is used; later restatements are ignored.
- **Trailing twelve months.** Flow items (sales, income, cash flow...) are TTM sums of quarters.
- **Groups.** S&P names carry GICS sector, industry and sub-industry labels. Other names get GICS-like labels from a SIC → GICS crosswalk learned on the S&P names. The Data page shows which classification the panel uses.
- **Liquidity universes** are rebalanced monthly on 63-day dollar volume.

The rest of the Data page

- **Active dataset** shows dates, stock and day counts, fields, universes, the in-sample (IS) and out-of-sample (OS) periods and the BRAIN-like window. **Use** switches between real and demo data.
- **Field coverage** shows, per field, the share of stock-days with a value over the last year. Low coverage explains weak or concentrated fundamental alphas.

4. Connecting to BRAIN

The connection is optional, but it is what turns a local candidate into a verified one.

What you get and what you do not

Available	Not available
Simulating any alpha on BRAIN: real Sharpe, fitness, turnover, returns, drawdown, margin, and every check	BRAIN's raw data (prices, fundamentals, analyst, news...). BRAIN does not allow downloading it
BRAIN's pre-submission check, including self- and production-correlation	Submitting alphas. The tool never submits
Your own alphas, their results and daily PnL	
The data-field catalog: ids, descriptions, coverage, and how many alphas use each field	

How to get access

There is **no API key**. BRAIN's API (api.worldquantbrain.com) accepts the same email and password as the BRAIN website. Any account that can sign in on the website can use it. Before automating anything, read the BRAIN terms and your consultant agreement on automated use and simulation limits.

Signing in

1. Open **BRAIN** in the sidebar.
2. Enter your BRAIN email and password.
3. Optional: tick **Remember in Windows Credential Manager**. The tool then signs in again by itself when the session expires (about every 4 hours). Instead, you can set the environment variables **BRAIN_EMAIL** and **BRAIN_PASSWORD** before starting the tool.
4. Press **Sign in**.
5. **Biometric check**. If your account uses biometric sign-in, a button **Open the verification page** appears. Open it, finish the check in your browser, then press **Complete sign-in**.

The header then shows **BRAIN connected**.

Where your secrets live. The session cookie is kept in `%LOCALAPPDATA%\AlphaFoundry\brain_session.json`, outside the OneDrive folder. A remembered password is kept in Windows Credential Manager (entry `AlphaFoundry-BRAIN`). Neither is ever written to the project folder, the database, job settings or logs. **Sign out** ends the session; **Forget saved password** also removes the stored password.

First things to do once connected

1. **Limits and defaults.** Set *Parallel simulations* to your account's concurrent limit (usually 3) and a *Daily budget* you are comfortable with (default 500).
2. **Data catalog → Import top datasets.** This makes BRAIN's field ids known to the tool.
3. **Your BRAIN alphas → Sync my alphas.** This imports your history, marks submitted alphas for the self-correlation check, and calibrates the local model.

Section 11 describes every part of the BRAIN page.

5. Finding your way around

Pages (sidebar)

Page	Use it to
Dashboard	See the library at a glance: top candidates by grade, recent jobs, family statistics, coverage of your submitted alphas
Studio	Write one alpha and test it thoroughly
Idea Forge	Turn an idea (text, paper, formula, code) into finished alphas
Miner	Run automatic searches in the background
Library	Browse, filter, compare, combine, export and send alphas to BRAIN
BRAIN	Sign in, import catalog and alphas, run BRAIN mining
BRAIN import	Paste BRAIN results by hand; calibration and bandit statistics
Explorer	Look up operators, data fields, templates and the 101 Alphas
Data	Build and inspect the dataset
Settings	Check thresholds, quality grade, simulation periods, performance, theme

Header

- **Page title** and **dataset chip** (source, number of stocks, date range). Hover over the chip to see the IS and OS periods.
- **BRAIN connected / BRAIN offline:** click it to open the BRAIN page.
- **N jobs running:** click it to open the Miner.
- **live / offline:** whether live job updates are streaming.
- **Ctrl+K:** command palette. Jump to any page, open a recent alpha, start a forge, toggle the theme.
- **Sun/moon:** light or dark theme.

Keyboard shortcuts

Keys	Where	Action
Ctrl+Enter	Studio	Simulate
Ctrl+Enter	Idea Forge	Forge the idea
Ctrl+S	Studio	Save to the Library
Ctrl+K	Anywhere	Command palette

6. Reading the results

Metrics

Metric	Meaning
Sharpe	Annualized mean over standard deviation of daily PnL
Fitness	Sharpe $\times \sqrt{(\text{Returns} / \max(\text{Turnover}, 12.5\%))}$, as on BRAIN. Below 12.5% turnover, cutting turnover no longer helps; only higher Sharpe or returns do
Turnover	Average daily share of the book that is traded
Returns	Annual PnL divided by half the book (book = \$20M)
Drawdown	Largest peak-to-trough fall of cumulative PnL, as a share of half the book
Margin	PnL per dollar traded, in basis points

Periods

Period	What it is
IS (in-sample)	From two years after the data start to two years before today. Selection and optimization use only this period
OS (out-of-sample holdout)	The most recent two years. It is reported, never optimized on, so it shows whether an edge survives
BRAIN-like window	The last five IS years, similar to the window BRAIN evaluates

Checks

These follow BRAIN's delay-1 rules. All thresholds are editable in Settings.

Check	Passes when
LOW_SHARPE	IS Sharpe ≥ 1.25 (delay 0: 2.0)
LOW_FITNESS	IS fitness ≥ 1.0 (delay 0: 1.3)
LOW / HIGH_TURNOVER	$1\% \leq \text{turnover} \leq 70\%$
CONCENTRATED_WEIGHT	No stock above 10% of the book
LOW_SUB_UNIVERSE_SHARPE	Sharpe in the smaller universe $\geq 0.75 \times \sqrt{(\text{sub size} / \text{universe size})} \times \text{Sharpe}$
SELF_CORRELATION	PnL correlation < 0.7 with every alpha marked submitted, unless the Sharpe is at least 10% higher
LOW_2Y_SHARPE (soft)	The last two IS years keep a Sharpe of at least 1.0

The tool also reports its own robustness warnings: OS degradation, yearly consistency, parameter stability and drawdown.

Quality grade

The grade is the single most important number in the tool. It answers the question "will this survive BRAIN?"

Grade	Meaning	What to do
A (BRAIN-ready)	Every hard check passes with a margin (Sharpe and fitness $\geq 1.25 \times$ BRAIN's minimum, turnover 3–45%), and the evidence holds (see below)	Send it to BRAIN
B	Passes every local check, but with thin margins or one robustness warning	Worth a BRAIN test; refine it first if you can
C	Near miss (one check fails narrowly), or passes locally with weak evidence	Refine it (Doctor, sweep, Re-engineer)
D	Fails	Drop it or rethink it

The evidence for grade A:

- the holdout Sharpe is at least 0.6 and at least 45% of the IS Sharpe;
- the BRAIN-like window Sharpe is above the minimum;
- at least 70% of the Sharpe survives when every window moves $\pm 25\%$;
- at least 70% of IS years are profitable;
- the sub-universe check passes;
- the expression has at most 40 nodes.

Once enough BRAIN results are imported, the calibrated BRAIN Sharpe estimate must also clear the minimum.

The **Quality** card lists the evidence that held and the reasons that cost grades. The grade letter appears in the Library, in job results and on the Forge results.

Pass likelihood and calibration

BRAIN pass likelihood is a logistic model of local metrics that estimates the chance BRAIN's checks pass. It starts from sensible defaults. After about 30 BRAIN results (imported or fetched through the API), it is refitted to your own data. After 5 or more results, the tool also maps local Sharpe to an **expected BRAIN Sharpe**. The fit and its quality are shown on the BRAIN import page.

7. Studio: write and test an alpha

The editor

- **Autocomplete** for operators and fields, **signature help** as you type, and **live error checking** with underlines.
- **Multi-statement alphas** are supported: assign variables with `name = expression;` and end with the alpha itself.

```
value = group_rank(ts_backfill(ebitda, 120) / enterprise_value, industry);
reversal = rank(-ts_delta(close, 5));
0.6 * value + 0.4 * reversal
```

- The **settings bar** sets region, universe, delay, decay, neutralization, truncation, pasteurization and NaN handling, as on BRAIN. Only USA can be simulated locally; other regions are for export and for BRAIN simulation.

Running

Press **Simulate** (Ctrl+Enter). A run takes roughly 0.1–3 seconds depending on the pool. The sub-universe test and window-stability probes follow a moment later; the Quality and Checks cards update when they arrive.

Panel	Shows
Metric strip	Sharpe, fitness, turnover, returns, drawdown, margin (IS, with OS below), BRAIN pass likelihood
Charts	PnL (IS and OS shaded), drawdown, rolling one-year Sharpe, turnover, long/short counts, yearly table
Quality	Grade, margins, evidence, reasons
Checks	Every BRAIN check and robustness gate, with values and limits
Robustness	Sharpe of the perturbed-window variants
Correlation	Most correlated Library alphas (click one to open it)
Attribution	PnL by sector, best and worst names
Description	A drafted BRAIN description: idea, data, operators

Improving an alpha

Button	What it does
Doctor	Proposes and simulates concrete fixes for each failing check: decay, smoothing, event gates, neutralization, size buckets, sign flip, back-filling... It ranks them by how many checks they fix. Click Apply to load one
Settings sweep	A decay × neutralization × truncation heatmap. The highlighted cell is the robust best (best neighbourhood), not the lucky peak
Re-engineer	A staged search that rebuilds a weak alpha: direction, shape, neutralization, horizon, turnover, conditioning, blending, settings. It gives a recipe and a holdout verdict. Set a time budget and targets, then start it; it runs as a background job

Saving and sending

- **Save** (Ctrl+S) stores the alpha, its full evaluation and its grade in the Library.
- **Copy** copies the expression. **BRAIN JSON** copies a simulation payload with settings.
- **Run on BRAIN** simulates it on BRAIN (needs the connection). The result appears in the Library.
- **History** lists your recent runs with their Sharpe, fitness and turnover. Click one to rerun it.

8. Idea Forge: from an idea to an alpha

The Forge turns an idea into finished, graded alphas. You describe **what** should work; it works out **how**.

What you can give it

Input	Example	How it is read
Plain English	<i>Stocks that drop sharply on heavy volume tend to bounce back within a week.</i>	Mechanism, direction, data, horizon, peer groups and event conditions are recognised
A written formula	<i>Signal = 12-month return skipping the last month, ranked within industry</i>	Compiled into <code>group_rank(ts_delay(ts_sum(returns, 231), 21), industry)</code>
Named ratios	<i>EV/EBITDA, earnings yield, book-to-market, ROE, gross margin, debt to equity, asset growth, FCF yield, low P/E</i>	Built from the right fields, with fundamentals back-filled

Input	Example	How it is read
101-Alphas or paper notation	<code>(-1 * correlation(rank(delta(log(volume), 2)), rank(((close - open) / open)), 6))</code>	Translated to BRAIN operators (<code>delta</code> → <code>ts_delta</code> , <code>correlation</code> → <code>ts_corr</code> , <code>IndNeutralize</code> → <code>group_neutralize</code> , <code>x^y</code> → <code>power</code> , <code>adv60</code> → <code>ts_mean(volume, 60)</code>)
pandas code	<code>mom = df['close'].pct_change(252).shift(21)</code> then <code>alpha = mom.rank(axis=1)</code>	Translated via Python's syntax tree (never executed) into a multi-statement alpha. A negative <code>shift</code> , which looks into the future, is refused
Fast Expressions	Any expression or program	Used as seeds
A document	PDF, Word, HTML, RTF, notebook, text	Text is extracted, references dropped, and the ~6 sentences that state the idea drive the search
JSON or YAML	<code>{"idea": "...", "fields": [...], "neutralization": "INDUSTRY"}</code>	Text, expressions, fields and settings are picked out
A list of ideas	Bulleted or numbered lines	Searched together; the Compose stage can combine them
Settings in the text	<i>neutralize by industry, decay 4, TOP1000</i>	Applied to the drafts

Type or paste into the box, or press **Open a file**. Try the example chips under the box to see how different inputs are read.

Checking how it read your idea

The **How the forge reads it** panel updates as you type (no simulation yet):

- **Read as:** the detected format, plus any settings found in the text.
- **Formulas built from your description:** the compiled expressions. Click one to open it in the Studio.
- **Key sentences** (documents): the sentences it used.
- **Mechanism and direction:** for example *Short-term reversal: long low recent return (as stated)*. Press **Change mechanisms** to correct it.
- **Data:** solid chips are data you named; dashed chips are typical inputs added for the mechanism; orange chips exist only on BRAIN.
- **Horizon:** short, medium or long. Click to override.
- **Conditions and seeds**, the **first drafts** it will try, and the **closest library templates**.

A *loose reading* means the text named neither the data nor what should happen next. Add both, for example *"... on heavy volume ... bounce back within a week"*.

Running a forge

Choose the effort and press **Forge alpha** (Ctrl+Enter).

Effort	Time budget	Use for
Quick	up to ~5 min	Checking an idea
Standard	up to ~12 min	Normal use
Deep	up to ~28 min	Ideas worth the time; more drafts and a two-island evolution

The stages run in the background, and the page shows live progress and a leaderboard:

Stage	What happens
Interpret	Reads the idea

Stage	What happens
Draft	Builds and screens recipe drafts, compiled formulas, seeds and matching templates. The top ones are also tested in the reverse direction
Combine	Blends and interactions across mechanisms, size tilts, event gates, smoothing, neutralization variants
Refine	Doctor fixes, the fitness shaper, and a robust settings sweep
Compose	Complex (multi-statement) alphas from the leaders: blends, tilts, regime switches, horizon ensembles
Evolve	Genetic programming seeded with the leaders, restricted to the idea's data
Polish	Full evaluation of the finalists: sub-universe, stability, every check, quality grade

You can leave the page; the forge keeps running. Open **BRAIN settings for the search** before starting to change region, universe, delay or other base settings.

The result

- **Champion alpha:**
 - its grade, status and metrics (IS with OS);
 - the evidence and the reasons behind the grade;
 - how it was forged, step by step;
 - a **hypothesis check** that says whether the data supports the direction you stated or favours the reverse.

Buttons: **Open in Studio**, **Copy**, **BRAIN JSON**, **Run on BRAIN**, **Library #id**.

- **Best simple (one-line) alpha** and **best complex (multi-statement) alpha**: both are always offered when both exist.
- **Most faithful variant**: shown when the champion leaves out data you named.
- **Runners-up**: decorrelated alternatives (PnL correlation below 0.7).
- **BRAIN-only variants**: drafts on data only BRAIN has, saved unscored. Test them with **Run on BRAIN**.

Everything is saved to the Library with the tags **forge** and **forge-<job id>**. The champion is tagged **forge-champion**.

Tips for good ideas

- Name the **data** (price, volume, earnings, debt, cash flow...), the **direction** (who outperforms) and the **horizon** (days, weeks, months).
- Say how stocks are compared: "within industry", "size-neutral".
- Mention conditions: "after earnings", "on heavy volume", "among low-volatility stocks".
- Paste formulas as they are; the translators handle paper notation and pandas.

9. Miner: automatic search

Choose a method card, set its options and press **Start**. Jobs run in the background. The right-hand panel shows live progress and the saved results, and has **Pause**, **Resume** and **Stop** buttons. Stopped jobs keep everything saved so far.

Method	What it does	Typical use
Auto-Mine	Quality-driven loop: bandit-chosen templates plus random grammar → screen → fitness shaping → Doctor on near misses → periodic genetic programming → periodic complex composites. Saves graded, decorrelated alphas	The default search
Complex composites	Multi-statement alphas from your best decorrelated Library alphas (or expressions you paste): weighted blends, volume-gated blends, tilts, regime switches, orthogonalized signals, horizon ensembles	After you have several good alphas
Genetic programming	NSGA-II evolution of expression trees (fitness × novelty × simplicity), with islands and successive halving	Refining a promising family
Template grid	Expands the idea-tagged template library over fields, windows, groups and settings	Systematic coverage of known ideas
Random grammar	Typed, unit-aware random expressions	Wide exploration
101 Alphas	The <i>101 Formulaic Alphas</i> as seeds	Price-volume starting points
Batch list	Evaluates and saves every expression you paste, one per line	Testing a list you already have
Settings optimizer	Robust decay × neutralization × truncation search for chosen Library alphas	Squeezing fitness out of an existing alpha
BRAIN-only ideas	Candidates on data that exists only on BRAIN	Feeding BRAIN tests
Re-engineer	The staged rebuild of one weak alpha	Rescuing an alpha
BRAIN mining	Opens the BRAIN page, where BRAIN judges every candidate	Verified results

Auto-Mine options

Option	Default	Meaning
Time limit	30 min	The job stops at this time ...
Stop after this many alphas of grade	10 of grade A	... or when this many alphas of the chosen grade are found
Save alphas graded at least	B	Use C to also keep near misses for later refinement
Save if Sharpe ≥	1.0	Floor for considering a candidate at all
Round size	24	Candidates generated per round
GP every N rounds	3	How often genetic programming refines the best material
Idea families	all	Restrict the search to chosen families
Build complex composites	on	Every few rounds, compose the good alphas found into multi-statement alphas
Also generate BRAIN-only ideas	off	Add BRAIN-only candidates at the end

The **base BRAIN settings** at the bottom of each method apply to all its candidates.

How long it takes

On a 2-core laptop, the demo panel evaluates about 30 candidates per second. The real S&P 1500 pool is several times slower, and the broad pool slower still. Plan 30–60 minutes for a meaningful Auto-Mine run on real data. Use **Settings** → **Performance** → **Miner worker processes** to match your cores.

10. Library: select, check and export

Every alpha you save, forge or mine lands here.

The table

Columns: selection, star, id, local status, **grade**, Sharpe, fitness, turnover, returns, OS Sharpe, sub-universe Sharpe, pass likelihood, highest correlation with another library alpha, idea family, BRAIN status and expression. Click a header to sort; the default sort is by quality. A **demo data** label in the BRAIN column marks alphas mined on the synthetic data. Ignore those.

Filters

Search (expression, notes, tags) · status · origin · idea · data category · **grade** (A; A or B; A to C) · **BRAIN status** (tested, ready to submit, passed, failed, not tested) · **dataset** (real or demo) · minimum Sharpe · starred · submitted · local only.

The alpha drawer

Click a row to open it. You see the expression and settings, then:

- **Open in Studio, Copy, BRAIN JSON, Run on BRAIN, Star, Re-engineer;**
- **Submitted on BRAIN:** marks the alpha so the SELF_CORRELATION check compares new alphas against it;
- notes and tags;
- the full result: metrics, charts, description, quality, checks and robustness;
- **Results on BRAIN:** real metrics, failing checks, self- and production-correlation, and a link to the alpha on the BRAIN website.

Bulk actions (select rows first)

Action	What it does
Export	BRAIN simulation payloads (JSON in batches of 10), plain expressions with settings comments, CSV with local metrics, or Markdown reports. Copy or download them
Correlation	PnL correlation heatmap of the selection, clustered. Red means redundant; BRAIN's limit is 0.7
Combine	A SuperAlpha-style PnL blend preview (equal, inverse-volatility or Sharpe weights)
Optimize settings	Starts the settings optimizer on the selection
Refine with GP	Starts genetic programming seeded with the selection
Compose complex	Builds multi-statement alphas from the selected alphas
Run on BRAIN	Simulates the selection on BRAIN
Star, Mark submitted, Unmark, Delete	Bookkeeping

11. The BRAIN page in detail

Connection

Status, session expiry, whether multi-simulation is allowed (up to 10 alphas per request), simulations used today, and the saved sign-in. **Sign out** and **Forget saved password** are here too.

Limits and defaults

Setting	Meaning
Parallel simulations	Simultaneous BRAIN simulations. Match your account's limit (usually 3)
Daily budget	The most simulations this tool may run per day. Jobs stop cleanly when it is reached
Multi-simulation	Auto uses it when your permissions allow
Region, universe, delay	Defaults for catalog imports and BRAIN mining
Run BRAIN's pre-submission check	For alphas that pass, also fetch self- and production-correlation. An alpha that clears everything is marked ready

Data catalog

Press **Import top datasets**, or tick datasets in the list first and press **Import N dataset(s)**. For each field, the import brings in the id, description, coverage, and how many alphas already use it. Afterwards:

- the Idea Forge finds BRAIN fields your idea describes (for example "analyst revisions" or "news sentiment") and drafts BRAIN-only alphas on them;
- BRAIN mining prefers well-covered fields that few alphas use, because crowded fields tend to fail correlation checks;
- the Explorer lists the fields.

Your BRAIN alphas

Sync my alphas imports your last 300 BRAIN alphas and their results. Submitted ones are marked submitted in the Library, and their PnL drives the local SELF_CORRELATION check. Every result recalibrates the pass model and rewards the idea families that really pass.

BRAIN mining

A closed loop in which BRAIN judges every candidate:

1. It takes your best untested Library alphas (grade A only, A or B, or A to C) and fresh ideas on imported catalog fields.
2. It simulates them on BRAIN within the budget.
3. It repairs near misses from BRAIN's own failing checks (decay, neutralization, smoothing, ranking...) and simulates the repairs, for up to *Repair rounds* rounds.
4. It records every result in the Library.

Option	Meaning
Simulation budget	Most simulations this run may use
Time limit	Wall-clock limit
Repair rounds	How many times near misses are repaired and re-simulated
Library alphas of grade	Which local alphas qualify

Option	Meaning
Best untested library alphas / ideas on imported catalog fields	The candidate sources

The **BRAIN jobs** list and panel show progress: simulated, passed, and ready counts.

12. BRAIN import (without the API)

If you prefer not to connect, or for results you already have:

1. Copy results from BRAIN as CSV, as JSON from your own scripts, or as text copied from the results panel.
2. Paste them into **Import BRAIN results** (or **Open file**), choose the format or leave it on auto-detect, and press **Import**. Tick **Mark all as submitted** for submitted alphas.
3. Rows are matched to Library alphas by canonical expression; new ones are created.

The page also shows:

- **Import BRAIN data-field catalog**: paste a field list (CSV or JSON), for example exported from BRAIN's data explorer;
- **Local vs BRAIN calibration**: local against BRAIN Sharpe, with the fitted mapping;
- **Pass rate by idea family** on BRAIN;
- **Search bandit**: which idea families and templates the miners currently favour, based on local grades and BRAIN outcomes.

13. Explorer

Tab	Contents
Operators	Every Fast Expression operator with signature, description and example. Operators marked BRAIN-only cannot be simulated locally
Data fields	Local fields and imported BRAIN fields, with descriptions and availability
Templates	The idea-tagged template library (about 90 templates, including multi-statement composites). Try in Studio opens an expansion, Mine this family starts the template grid. Add / update a user template saves your own to <code>runtime/templates_user.yaml</code>
101 Alphas	The translated formulas from <i>101 Formulaic Alphas</i> . Verify them against the paper before relying on them

14. Settings

Card	Settings
Submission check thresholds	Delay-1 and delay-0 Sharpe and fitness minimums, turnover limits, max weight, sub-universe factor, self-correlation limit, Sharpe improvement and years, soft two-year Sharpe, local robustness gates. Quality grade A section: Sharpe and fitness margins, turnover band, holdout Sharpe and ratio, stability, maximum nodes
Simulation	Years of warm-up before IS, OS holdout length, BRAIN window length, book size, returns basis, history start
Performance	Miner worker processes, server and worker cache sizes
Appearance	Theme

Changes to periods or the dataset reload the data automatically. Check thresholds are saved to `runtime/checks_user.yaml`; the defaults stay in `backend/alphafoundry/catalog/checks.yaml`.

Tighten the quality margins if too many grade A alphas fail on BRAIN. Loosen them slightly if BRAIN passes alphas the tool rates B.

15. Recommended workflows

A. Your first week

1. **Data:** build the Broad US pool (section 3).
2. **BRAIN:** sign in, import the catalog, sync your alphas (section 4).
3. **Miner** → **Auto-Mine:** 30–60 minutes, target 10 grade A alphas.
4. **Library:** filter *Grade A or B*, sort by quality, select the top 10–20 → **Correlation** to drop near-duplicates → **Run on BRAIN**.
5. **Library:** filter *BRAIN: ready to submit* and submit the best on the BRAIN website.
6. **BRAIN** → **Sync my alphas**, so self-correlation and calibration include what you submitted.

B. Turn a paper or a formula into alphas

1. **Idea Forge** → **Open a file** (PDF or Word), or paste the formula or code.
2. Check the reading: format, compiled formulas, key sentences, mechanism, data. Correct the mechanism or horizon if needed.
3. **Standard** effort → **Forge alpha**.
4. Compare the champion, the best simple and the best complex alpha. Read the hypothesis check.
5. **Run on BRAIN** on the one with the best grade. Test the BRAIN-only variants too.

C. Fix an alpha that failed on BRAIN

1. Open it in the **Studio**.
2. Press **Doctor**. Apply the fix that clears the check BRAIN failed (see section 16).
3. Run the **Settings sweep** and take the robust cell.
4. If it is still weak, press **Re-engineer** with a 5–10 minute budget.
5. Save the result and **Run on BRAIN**.

D. Build a complex alpha from your best ones

1. **Library:** filter grade A or B and select 3–8 alphas of different families that are not correlated (check with **Correlation**).
2. **Compose complex.** The job writes blends, tilts, regime switches and horizon ensembles and saves the good ones, tagged `complex`.
3. Compare each composite with its best part: it should add Sharpe or stability, not just complexity.

E. Stay diverse (avoid SELF_CORRELATION)

- Mark submitted alphas as submitted, or sync them from BRAIN.
- Filter by idea family and mine families you have few alphas in (the Dashboard shows the spread).
- Prefer less-used BRAIN fields (BRAIN mining does this automatically).
- Change the dataset first, then the peer group, then the horizon. These change the PnL path most.

F. A daily routine

1. **Data** → **Update data** (a few minutes).
2. Start an **Auto-Mine** or a **BRAIN mining** run in the morning.
3. Review new grade A alphas and BRAIN-ready alphas in the Library at the end of the day.

16. Fixing failed checks: a cheat sheet

Failed check	What to try in the tool
LOW_FITNESS with high turnover	Increase decay (6–16); <code>ts_decay_linear(x, 5–10)</code> ; a volume-event gate <code>trade_when(volume > adv20, x, -1)</code> ; <code>hump(x, 0.02–0.05)</code> . Doctor and the fitness shaper try all of these
LOW_FITNESS with turnover already below 12.5%	Only a higher Sharpe or returns help: peer-relative ranking (<code>group_rank</code>), a sharper weight profile (winsorized <code>zscore</code> instead of <code>rank</code>), a better neutralization
LOW_SHARPE	Rank or z-score and winsorize; test other neutralizations; peer-relative versions; condition on volume or volatility; check the reverse sign (the Forge's hypothesis check does this)
HIGH_TURNOVER	As for LOW_FITNESS with high turnover, plus longer windows
LOW_TURNOVER	Decay 0; shorter windows; use changes (<code>ts_delta</code> , <code>ts_zscore</code> , <code>ts_rank</code>) instead of static levels
CONCENTRATED_WEIGHT	Wrap in <code>rank()</code> ; <code>winsorize(x, std=3)</code> ; truncation 0.05; <code>ts_backfill</code> sparse fundamentals
LOW_SUB_UNIVERSE_SHARPE	The edge lives in small caps: rank instead of raw ratios; <code>group_neutralize(x, bucket(rank(cap), range="0.1,1,0.1"))</code> ; sub-industry neutralization
SELF_CORRELATION	Different dataset or field; different peer group; double or halve the horizon; different operator family; <code>regression_neut</code> against the correlated alpha; a complex composite with a new component
Grade B/C for weak evidence (holdout, stability)	Standard windows (Doctor → Standard windows); simplify the expression; prefer fewer nodes; horizon ensembles (Compose)

17. Command line

Run these from a terminal in the **backend** folder.

```
uv run alphafoundry serve --open          # start the app (options: --host, --port)
uv run alphafoundry warmup                # recompile the engine kernels (after code updates)
uv run alphafoundry build-data            # build or update real data (pool from Settings)
uv run alphafoundry build-data --max-tickers 200 # quick trial build
uv run alphafoundry demo                  # rebuild the synthetic demo dataset
uv run pytest                             # run the test suite
uv run python scripts/bench.py            # responsiveness benchmark on the active dataset
```

UI development: `cd frontend` then `npm run dev` (serves on port 5173 and forwards to the backend on 8765).

18. Files, backup and reset

Path	Contents
<code>runtime/alphafoundry.db</code>	The Library, results, jobs, BRAIN results, calibration and bandit (SQLite)
<code>runtime/panels/</code>	The real dataset (large; can be rebuilt)
<code>runtime/panels_demo/</code>	The demo dataset (rebuilt automatically)
<code>runtime/raw/</code>	Download caches for Yahoo, SEC and Wikipedia (make updates fast)
<code>runtime/settings.json</code>	App settings
<code>runtime/checks_user.yaml</code>	Your check and quality thresholds
<code>runtime/templates_user.yaml</code>	Your own templates
<code>runtime/fields_user.json</code>	Imported BRAIN field catalog
<code>runtime/logs/</code>	Server log
<code>%LOCALAPPDATA%\AlphaFoundry\</code>	BRAIN session cookie (outside OneDrive)
<code>docs/brain-api.md</code>	BRAIN connection reference

- **Back up** `runtime/alphafoundry.db` and the three `*_user.*` files. The datasets can always be rebuilt.
- **Reset everything:** stop the tool and delete the `runtime` folder. The next start recreates it with the demo data.
- **Reset only the Library:** stop the tool and delete `runtime/alphafoundry.db*`.
- **OneDrive:** the project folder is synced. If you see file-lock errors or slowness, pause sync during data builds and long mining runs.

19. Troubleshooting

Problem	Solution
"The active dataset is the synthetic DEMO panel..." when starting a job	Build real data (section 3). To try the tool anyway, tick <i>Run on demo data anyway</i> ; the results will not transfer to BRAIN
SEC or Wikipedia returns 403	Enter a real contact email on the Data page
Yahoo returns 429	The downloader waits and retries. Run the build again later; cached tickers are skipped
The build stops partway	Run it again. Everything downloaded so far is cached
First simulation or job is slow	The engine compiles on first use. start.bat does this once; rerun uv run alphafoundry warmup after code updates
Port 8765 is busy	The server uses the next free port and prints it
"BRAIN rejected the email or password"	Check them by signing in on the BRAIN website
Biometric link keeps coming back	Finish the check in the browser before pressing Complete sign-in; the link expires after a few minutes
"BRAIN daily simulation limit reached"	BRAIN's own daily limit. Wait for the reset; results so far are kept
"Today's BRAIN simulation budget is used up"	The tool's own cap. Raise <i>Daily budget</i> on the BRAIN page
A field is unknown on BRAIN	The local name may not exist in that region or universe. Import the catalog and check the field in the Explorer
Grade A alphas still fail on BRAIN	Import or sync the results so calibration learns, and raise the quality margins in Settings. Make sure you use the Broad US pool
No grade A alphas are found	Run longer, try more families, keep C grades (<i>Save alphas graded at least: C</i>) and refine them, or lower the margins slightly
A scanned PDF gives no text	The PDF has no text layer. Run OCR on it first, or paste the key passages

20. Glossary

Term	Meaning
Alpha	A Fast Expression that assigns each stock a weight every day
Book	The total position size (\$20M), half long and half short after neutralization
Decay	Linear averaging of the alpha over the last n days; lowers turnover
Delay	Days between the data and the trade: 1 on BRAIN's standard setting
Fast Expression	BRAIN's alpha language
Fitness	$\text{Sharpe} \times \sqrt{(\text{returns} / \max(\text{turnover}, 12.5\%))}$
Grade	Alpha Foundry's quality rating: A (BRAIN-ready) to D
IS / OS	In-sample (used for selection) and out-of-sample holdout (reported only)
Neutralization	Removing market, sector, industry or sub-industry exposure by demeaning within groups
Pool	The set of stocks in the local dataset (Broad US or S&P 1500)

Term	Meaning
Sub-universe	A smaller, more liquid universe (for example TOP1000 inside TOP3000) used for a robustness check
Truncation	Cap on any single stock's share of the book
Universe	The tradable stocks on a given day, chosen by liquidity (TOP3000, TOP1000...)

21. Limits to keep in mind

- **Local results are a proxy.** The free data differs from BRAIN's: prices from Yahoo, fundamentals from SEC filings, VWAP approximated by $(\text{high} + \text{low} + \text{close}) / 3$, and today's listed stocks only, so there is survivorship bias. BRAIN is the ground truth; the quality grade and calibration are there to bridge the gap.
- **Delay-0 results are optimistic** locally, because trades use the same close the data comes from.
- **Only USA can be simulated locally.** Other regions and risk-model neutralizations work through BRAIN (Run on BRAIN, BRAIN mining) and exports.
- **The BRAIN API endpoints** are the ones BRAIN's website and community tools use. They are not an official, versioned API, and BRAIN may change them. If a call starts failing after a BRAIN update, the error names the endpoint.
- **Some thresholds are unverified** against official BRAIN documentation: the delay-0 minimums, the sub-universe factor, the two-year check and the returns basis. They are editable in Settings.
- **Keep submitting selectively.** Test many, submit few, and diversify across data, horizons and ideas.